



$$\begin{aligned}
 y_{Mixer}(t) &= x_{DSB}(t) \cdot \cos(\omega_c t) & \circ \text{---} \bullet & Y_{Mixer}(\omega) &= \frac{A_c}{2} \cdot M(\omega) + \frac{A_c}{4} \cdot [M(\omega - 2\omega_c) + M(\omega + 2\omega_c)] \\
 y_{TP}(t) &= T_{TP} [y_{Mixer}(t)] & \circ \text{---} \bullet & Y_{TP}(\omega) &= \frac{A_c}{2} \cdot M(\omega)
 \end{aligned}$$